

PRN212 – Assignment

Part 1: Delegate (3.0 Points)

Exercise 1: Basic Delegate

1. Create a **delegate** StringOperation that takes a string and returns encrypted message.
2. Write a method encryption and use the delegate to call this method.

Encryption is one of the ways to protect data event if it is transmitted over the Internet. A new encryption method is used to encode integers according to the following principle:

For example,

Digit	0	1	2	3	4	5	6	7	8	9
Encrypt value	f	p	t	u	c	a	n	T	h	o

Event if the given encrypted message is “tfta”, so the integer must be 2025.

Event if the given encrypted message is “cutpfnTho”, so the integer must be 432106789.

Input

Only one line contains the encrypted message of an integer N. The maximum length of message is 18 digit characters.

Output

Please print out the value of integer N.

Input	Output
tftc	2024
cutpfnTho	432106789

Exercise 2: Delegate as a Method Parameter

1. Write a method ExecuteOperation that takes a delegate MathOperation and two integers.
2. Pass the Add, Subtract, and Multiply methods into ExecuteOperation.

Part 2: Event (2.0 Points)

Exercise 3: Basic Event Usage

1. Create a Notifier class with an event OnNotify.
2. When calling the Notify() method, trigger the OnNotify event and display a message.

Exercise 4: Event with Data Parameters

1. Create a Sensor class with an event OnThresholdReached.
2. When the temperature exceeds 50°C, trigger the event.
3. Write an AlertSystem class to subscribe to the event and display an alert.

Part 3: LINQ (2.0 Points)

Exercise 5: Query a List of Objects

1. Create a List<Employee> with properties Name, Salary, Department.
2. Use LINQ to filter employees with a salary above 5000.

Exercise 6: LINQ with Multiple Conditions

1. Filter employees in the "IT" department with a salary above 7000.
2. Sort the list by salary in descending order.

Part 4: Combined - Delegate, Event & LINQ (3.0 Points)

Exercise 7: Bank Account Monitoring Application

1. Create a BankAccount class with an event OnBalanceChanged.
2. When the balance changes, trigger the event and display the new balance.
3. Use LINQ to retrieve transactions with a value above 1000.

Exercise 8: Order Management with LINQ

1. Create a List<Order> with properties OrderID, CustomerName, TotalAmount.
2. Use LINQ to:
 - Filter orders with a total amount above 500.
 - Calculate the total revenue of all orders.
 - Find the order with the highest value.